

Question Paper

Exam Date & Time: 13-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2025

ADVANCED MACHINE LEARNING [CSE- 5022]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast supervised and unsupervised learning in terms of data requirements and learning objectives. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Evaluate the distance if $k=6$ for the given vector $P = (15, 48, 29, 63, 90, 77, 41, 58, 12, 39)$ and $Q = (45, 72, 80, 41, 60, 92, 18, 54, 21, 33)$. (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Outline the examples of Condensed Perception Learning. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 2) Evaluate the effectiveness of the following methods in improving the efficiency of the Nearest Neighbor classifier: (4)
- A)
- a. Branch and Bound
 - b. Hypercube Pruning (Cube)
 - c. Projection-based Filtering
- Ordered Partitioning
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Compare and contrast Minimal distance classifiers and Centroid Classifier. (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Explain K-means clustering with suitable examples. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Apply the Leader Clustering Algorithm to the following data points with a threshold distance (T) = 3.0. Use Euclidean distance for clustering. List the leaders and assign each data point to its corresponding leader. (4)
- A)
- Data Points: A(1, 2), B(2, 1), C(4, 5), D(7, 8), E(5, 4), F(8, 8)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) (4)

For the given initial population Evaluate the probabilities of selection using the fitness proportionate method (Roulette Wheel Selection) and determine which individuals are most likely to be selected for reproduction.

$$s_1 = 1011010111 \quad f(s_1) = 6$$

$$s_2 = 1100111010 \quad f(s_2) = 5$$

$$s_3 = 1001110111 \quad f(s_3) = 7$$

$$s_4 = 1110010001 \quad f(s_4) = 4$$

$$s_5 = 1011111110 \quad f(s_5) = 8$$

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- C) Make use of the data provided and the entropy values, determine the Information Gain for the attribute "Temp" when building a decision tree to predict whether to "Play Tennis" or "Not" (2)

Day	Temp	Humidity	Wind	Paly Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

$$S_{Sunny} = [2+, 3-] \quad Entropy(S_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{Hot} \leftarrow [0+, 2-] \quad Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} \leftarrow [1+, 1-] \quad Entropy(S_{Mild}) = 1.0$$

$$S_{Cool} \leftarrow [1+, 0-] \quad Entropy(S_{Cool}) = 0.0$$

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Determine the winning neuron and the weights of the winning neuron using learning rate $\alpha = 0.5$ for (4)

- A) the given Kohonen Self-Organizing Map with 2 input features and 3 neurons in the output layer.
The initial weight vectors for the neurons are:
W1=[0.2,0.6]
W2=[0.8,0.4]
W3=[0.5,0.9]
A new input vector X=[0.6,0.5] presented.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) The RBF network has: (4)
One RBF neuron with center C=[1.0, 2.0]
The RBF uses Gaussian activation:

Spread (σ) = 1.0

Weight w=0.9

Bias b=0.1

Output of the RBFN: $y(x)=w \cdot (x)+b$

A new input vector is x= [2.0, 3.0].

Determine the activation of the RBF neuron, final output of the network and evaluate whether the network output crosses the threshold (assume threshold = 0.5) for classification.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Choose the reasons why even if we have an extremely powerful model, we cannot always achieve zero error (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) **Given two classes ω_1 and ω_2 with the following properties:** (4)

$P(\omega_1) = 0.6, P(\omega_2) = 0.4$

- A) **The class-conditional probability density functions are:**

$p(x | \omega_1) = 1/\sqrt{2\pi} \cdot e^{-(x-2)^2/2}$

$p(x | \omega_2) = 1/\sqrt{2\pi} \cdot e^{-(x-5)^2/2}$

Analyze by using the concept of minimum error-rate classification to determine the decision boundary and classify a sample with x = 3.0.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Explain different types of Ensemble learning models. (4)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- C) Choose the different components of multi-expert system. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination CSE- 5022:ADVANCED MACHINE LEARNING
- Apr-May 2025

#	Course Outcome	Description	Marks
1	1	CO 1	4

2	2	CO 2	16
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10

Total 50

* **NA** - CO Not Selected

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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	4(8%)
3	Application	2	4(8%)
4	Analysis	4	14(28%)
5	Create	7	28(56%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)